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GROUP PROJECT TITLE : PARKINSON'S DISEASE DETECTION SYSTEM

DOMAIN : MEDICAL

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1.0 INTRODUCTION

The "Parkinson's Disease Detection" dataset provides an extensive overview of various physiological and clinical indicators associated with Parkinson's disease (PD), including voice assessments for motor and non-motor symptoms. The symptoms of Parkinson's disease can vary for each person based on their body's reaction. In the early stages of Parkinson's disease, the ability to make facials. When walking, the expressions may be lacking. These symptoms include tremors, which involve shaking in a limb, often the hand and fingers arms may remain still. Additionally, speech can become slurred, soft, quick, or hesitant before talking. Speech changes, such as a monotone quality replacing the usual speech pattern are among the main symptoms studied in this research. Over time, the symptoms of Parkinson's disease worsen. This study utilizes the Feedforward Neural Network method to predict the occurrence and progression of Parkinson's disease based on these performance characteristics.

Back then, medical professionals would examine a patient's neurological history and perform a standard motor skill assessment in order to diagnose Parkinson's disease traditionally. Achieving an accurate diagnosis can be difficult since there is no reliable laboratory test for Parkinson's disease (PD), particularly in the early stages when motor symptoms are modest. Patients must attend several clinic appointments over time in order to track the disease's development. Then, since the speech and vocal abnormalities are common in people with Parkinson’s Disease, the voice recordings can be a valuable and non-invasive diagnostic tool for future use and development.

By applying machine learning algorithms to a collection of speech recordings, the diagnostic procedure for Parkinson's disease (PD) can be streamlined, potentially leading to faster diagnoses. This non-invasive method allows medical experts to detect early indicators of PD without the need for invasive treatments. The dataset used in this study comprises various biomedical voice measurements from 31 individuals, 23 of whom have Parkinson's disease. It includes 195 voice recordings, with each column representing a specific voice measure. Utilizing machine learning for speech analysis can facilitate the tracking of the disease's progression over time, providing doctors with real-time data and enabling more personalized treatment plans. Additionally, by lowering the frequency of in-person clinic appointments, this strategy can lessen the workload for medical professionals as well as patients.It creates opportunities for remote monitoring, making it available to patients in underserved or rural locations.

1.1 PROBLEM STATEMENT

Parkinson's disease (PD) is a degenerative neurological condition that affects patients' quality of life severely. It is characterized by the loss of dopamine-producing neurons in the brain, which results in a variety of motor and non-motor symptoms. These symptoms include bradykinesia (slowness of movement), tremors, muscle rigidity, and noticeable speech difficulties include hypophonia (lowered volume), dysarthria (difficulty articulating sounds), and monotone voice (limited pitch range). Moreover, PD may result in mood swings, cognitive impairments, and a higher chance of dementia.

Because there are no reliable laboratory tests and clinical assessments are subjective, it can be difficult to diagnose Parkinson's disease accurately, especially in the early stages. The majority of current diagnostic techniques rely on neurologists' evaluations of motor abilities and patient histories, which are frequently inadequate in the early stages of the disease. Additionally, frequent clinical visits are necessary for routine monitoring of illness progression, which puts a significant strain on patients and healthcare systems.

By applying machine learning algorithms to voice data, the dataset on the UCI Machine Learning Repository offers a chance to investigate a non-invasive diagnostic substitute. A model like this could greatly improve patient care and disease management by enabling early diagnosis and ongoing monitoring of Parkinson's disease.

1.2 OBJECTIVES

The Parkinson's Disease Detection system aims to distinguish between healthy individuals and those with Parkinson's disease. This primary objective is achieved by systematically assessing the classification performance across various vocal cord characteristics. By focusing on voice analysis, the system leverages subtle changes in speech patterns that are indicative of Parkinson's. Additionally, the integration of machine learning algorithms enhances the system's ability to accurately identify these changes. As a result, early detection becomes more feasible, allowing for timely medical intervention and improved patient outcomes.

Furthermore, other than classification, the secondary objective of the project is to recognize the range of biomedical voice measurements from the cases. The goal of this work is to find patterns in speech data that may indicate Parkinson's illness. Voice measures are important because they might reveal problems with motor control related to the illness. Through the examination of these patterns, experts can get a better understanding of the ways in which Parkinson's disease impacts speech and create more accurate diagnostic standards.In the end, the research findings from this project may help with the development of targeted treatments and interventions to control and lessen Parkinson's disease symptoms.

Next, the project’s objective is also to build a parkinson detection system using the Feedforward Neural Network algorithm. This involves creating a prediction model that makes use of a number of characteristics obtained from biological voice measurements. In order to guarantee that the dataset is clean and prepared for analysis, the procedure starts with extensive data preparation. We can ensure that the model generalizes successfully by training the FNN on a subset of the data and assessing its performance on unseen data by dividing the data into training and test sets.

The final objectives are to assess the model system through the experimentation of several hyperparameters, including the amount of the training data, the number of epochs, activation, optimizer, and the learning rate. This means developing multiple configurations and using cross-validation to acquire dependable performance estimates for every configuration. Through varying learning rates and epoch counts during training and performance validation, we may adjust the hyperparameters of the model to maximize its precision and resilience. This thorough assessment guarantees that the finished model is optimized and functions effectively when applied to fresh, untested data.

1.3 SCOPE

The Parkinson's disease detection system is designed to identify unusual voice patterns from various individuals and aims to detect any symptoms indicative of Parkinson's disease. First of all, the primary scope of this project is to develop a more accurate and reliable model for early disease detection. This project is designed to identify Parkinson's disease at its earliest stage, even before the clinical systems become apparent to the person. This can be done by developing methods to detect subtle voice changes that may indicate Parkinson's disease, as it can occur years before motor symptoms. Early detection can also be a timely medical intervention where neuroprotective strategies and treatments can slow disease progression, with the help of early detection of Parkinson disease.

Furthermore, the scope of this system is to create an automated system that implements machine learning algorithms to identify Parkinson's disease from biomedical voice recordings. FeedForward Neural Network (FNN) algorithm is used as it can detect patterns, especially in speech analysis. The system can also extract vital voice features that are found in parkinson's disease symptoms from recordings such as frequency, amplitude, jitter, and shimmer. From this, this system can create a training model where voice recordings are distinguished as either Parkinson's positive or negative. To make sure the model can generalize effectively to new, unseen data, it is verified using a variety of performance metrics, including accuracy, precision, recall, and F1 score.

Moreover, the system's scope is in clinical and practical implications where patient care and medical practice can be improved. Diagnosis standards can be enhanced as the system helps medical experts understand how Parkinson's affects speech. Hence, more accurate diagnoses can be developed. As the system automates the detection of Parkinson disease, the medical experts workload is reduced as the automated system helps reduce the burden of medical experts and healthcare officials. Thus, this allows them to focus more on treatment and care of the Parkinson patients, instead of the detection of the disease. The system also provides real-time monitoring where doctors are able to tailor treatment strategies to meet the specific needs of each patient. For remote patients, the system allows for continuous remote monitoring, resulting in reducing the need for frequent clinic visits and making healthcare more accessible to everyone.

2.0 DATA ANALYSIS

2.1 DATASET INFORMATION

The Parkinson Disease Detection dataset was obtained from Kaggle's website. A variety of biological speech measurements from 31 individuals, 23 of whom have Parkinson's disease (PD), make up this dataset. Each row in the table corresponds to one of the 195 voice recordings made by these people ("name" column), and each column in the table represents a specific voice measure. The "status" column, which is set to 0 for healthy and 1 for PD, indicates that the primary goal of the data is to distinguish between healthy and PD patients.

2.2 DATASET DESCRIPTION

Number of Instances	195
Number of Attributes	24
Number of Individuals	31 (23 with PD and 8 healthy)
Target Variable	"status" (binary: 0 for healthy, 1 for PD)
Associated Task	Classification

2.3 SOURCE OF DATA

- jaina. (2024). *Parkinson Disease Detection*. Kaggle.com. https://www.kaggle.com/datasets/jainaru/parkinson-disease-detection

3.0 FNN ARCHITECTURE

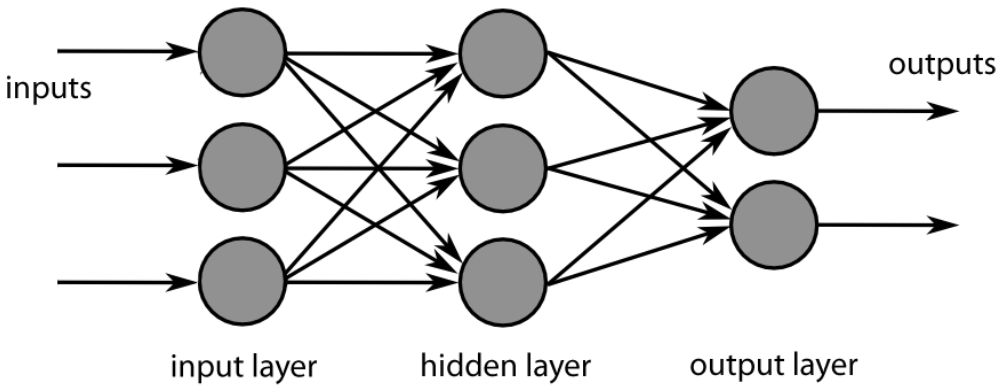


Figure 3.0.1 The Structure of FNN Architecture

In this project, we used the FeedForward Neural Network (FNN). Since the dataset has 24 features, the input layer of the FNN will have 24 nodes. The output layer is responsible for the classification task. Since this is a binary classification problem where we must determine between healthy people and patients with Parkinson's Disease, the output layer will have a single node, producing a probability value between 0 and 1.

3.1. HYPERPARAMETERS

3.1.1. Hyperparameters in Network Structure

BASE

1. Number of Hidden Layer and Units

- We have a total of 32 hidden layers for the first hidden layer and 16 hidden layers for the second hidden layer with a size of ??

2. Dropout

- We use dropout 0.2 (20%)

3. Activation Function

- The activation functions we use are ReLU, and Sigmoid activation functions.

3.1.2. Hyperparameters in Training Algorithm
BEST

1. Learning Rate

- The learning rate that we use is 0.004

2. Number of Epochs

- The number of times the learning algorithm will go through the entire training dataset. The number of epochs is increased until the accuracy is validated. The number of epochs we use is 159 and the final highest accuracy is 92.31%.

3. Batch Size

- The batch size we use is 32.

4.0 EXPERIMENTS

4.1. PARAMETER

4.1.1. Constant Parameter

Activation

The constant parameters that we use in action are functions. The first function is Rectified Linear Unit (ReLu), which is used in hidden layers. Next function is sigmoid, which is used for the output layer. For scenarios involving classification, this activation is quite helpful.

Layer

For hidden layer parameter we used:

Dropout: In order to prevent overfitting in neural networks, dropout is a regularization strategy.

Dence: a layer that connected to the every neuron in the previous layer

4.1.2. Change in Parameter

Throughout the experiment, the optimizers' parameters change.
This is the optimizer that we apply:

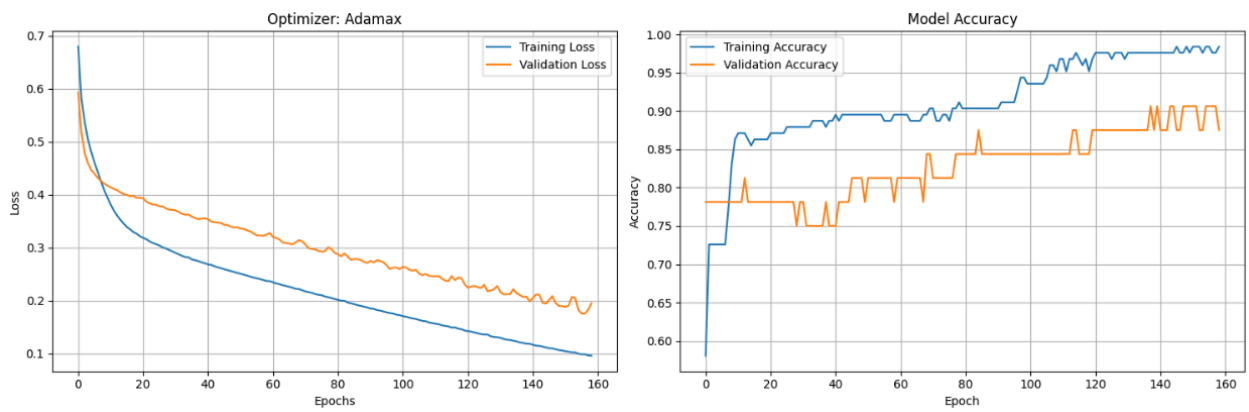
- **Adam**: It adapts the learning rate for each parameter, making it highly effective for a wide range of problems
- Adamax: Based on the infinite norm, Adamax is an Adam optimizer variant.
- SGD: It is the simplest and most widely used optimization algorithm.
- FTRL: algorithm for optimisation created for linear models with big feature spaces.
- Nadam: It aims to improve the convergence speed and performance of the model
- AdaGrad: modifies each parameter's learning rate according to past gradient magnitudes.
- RMSprop: It divides the learning rate by the square root of the moving average of squared gradients for each parameter, which it keeps track of.
- AdaDelta: Throughout training, it continually adjusts the learning rates without requiring a manually defined global learning rate.

4.2. ACCURACY RESULTS

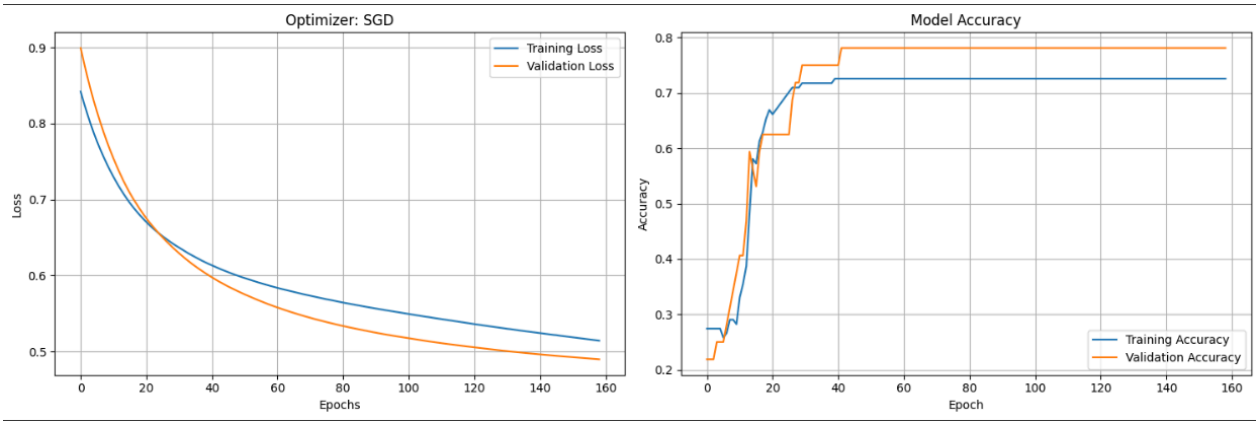
We tested with various optimizers on the model to identify the most effective one. With 159 epochs as a constant, different optimizers would yield different accuracy values. Following are the accuracy results for each of them:

4.2.1. Optimizer ‘Adam’

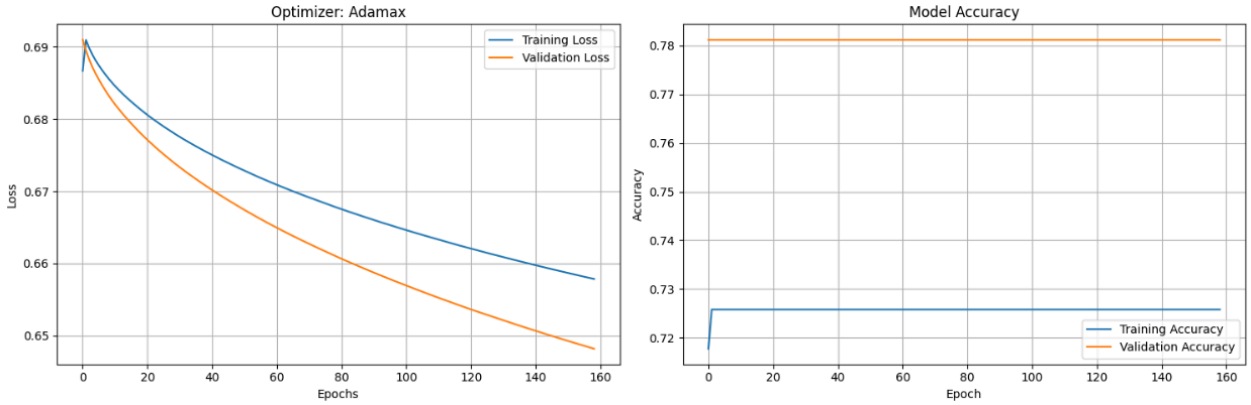
4.2.2. Optimizer ‘Adamax’



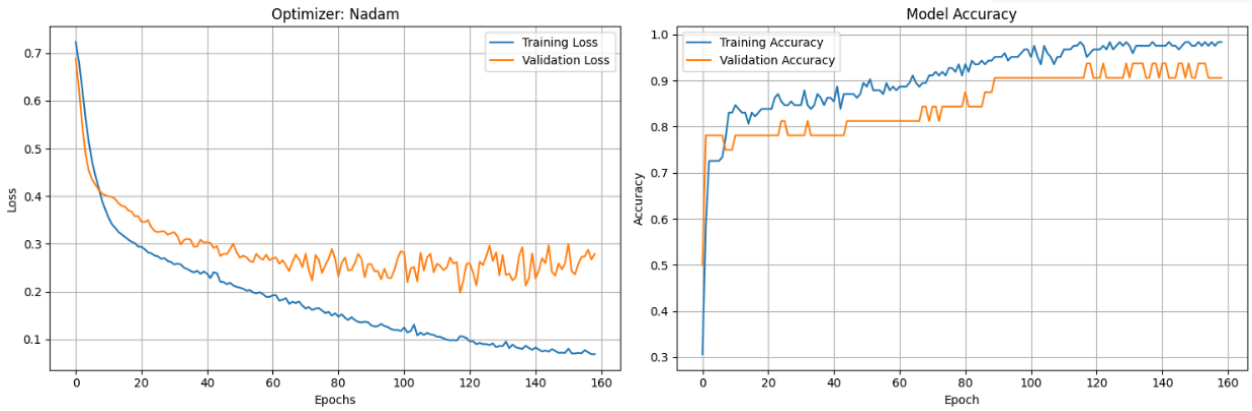
4.2.3. Optimizer ‘SGD’



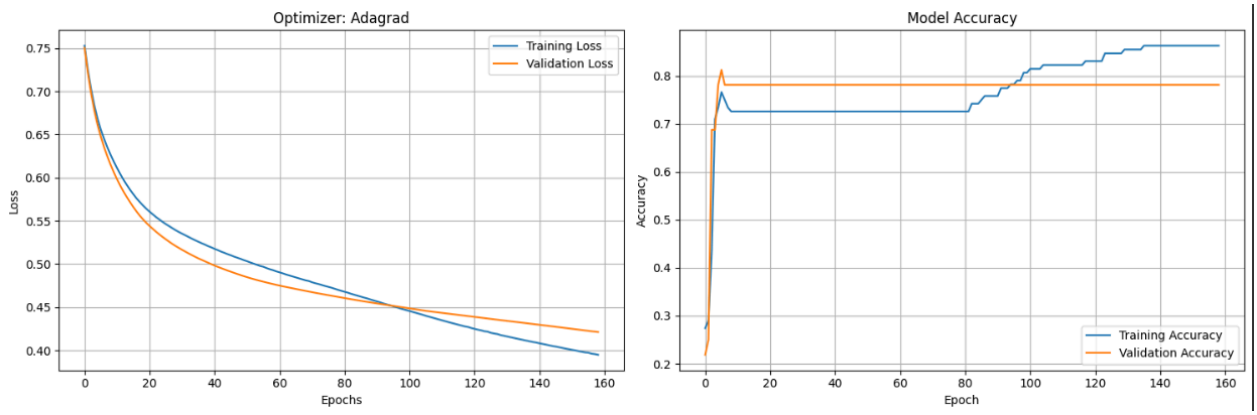
4.2.4. Optimizer ‘Ftrl’



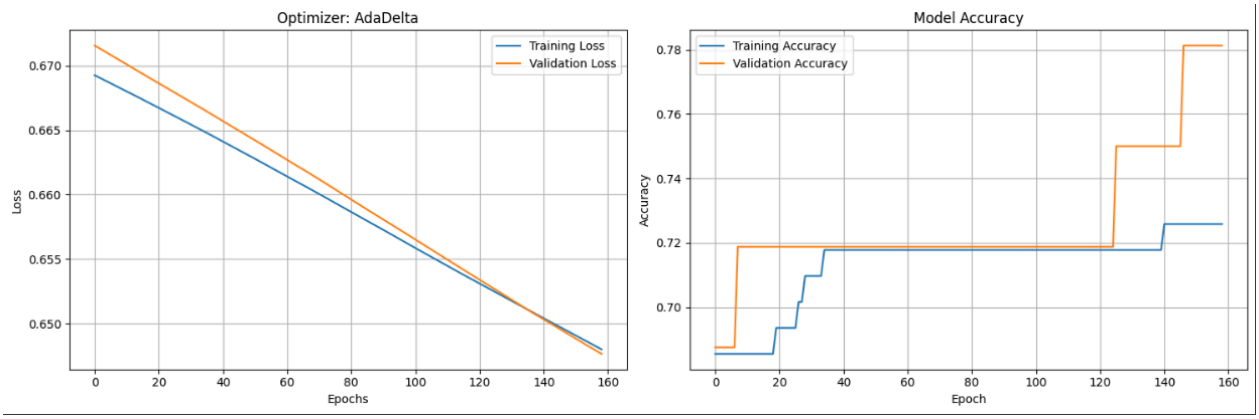
4.2.5. Optimizer ‘Nadam’



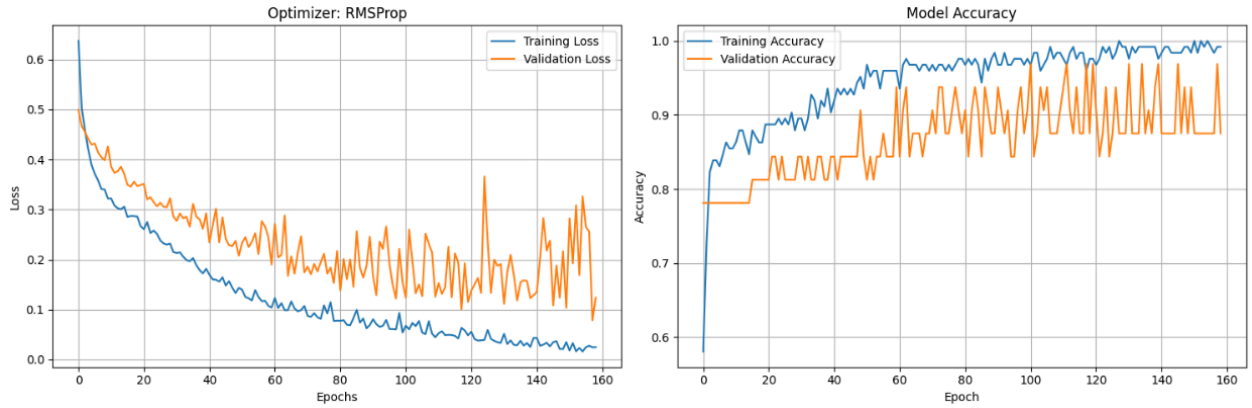
4.2.6. Optimizer ‘AdaGrad’



4.2.7. Optimizer ‘AdaDelta’



4.2.8. Optimizer ‘RMSProp’



In summary, the accuracy of each optimizer is as follows:

Optimizers	**Accuracy**
Adam	89.74%
Adamax	92.31%
SGD	82.05%
Ftrl	82.05%
Nadam	92.31%
AdaGrad	89.74%
AdaDelta	82.05%
RMSProp	89.74%

5.0 RESULTS

5.1. RESPONSE TIME

- We used Phyton to write our scripts for the dataset training and testing.
- The final accuracy rate obtained is 92.31% after using ‘Nadam’ and ‘Adamax’ optimizer.

5.2. MEMORY USAGE

- We used Google Collab to run the codes. Google Colab supports Python and allows users to install additional libraries. Google Colab also integrates with Google Drive, making it easy to load data from and save results to our Drive. This integration simplifies data management and sharing as we store our data in Drive and run it in Google Collab.
- The dataset’s size memory is only 16KB on kaggle.

5.3. NUMBER OF EPOCHS

- With 159 numbers of epochs, we acquired an accuracy rate up to 92.31%.

5.4. ACCURACY OF OPTIMIZERS

- Based on our training model, the highest accuracy is the ‘Nadam’ and ‘Adamax’ optimizer with 92.31%. The second highest is the ‘Adam’, ‘AdaGrad’ and ‘RMSProp’ optimizer with 89.74%. Followed by an accuracy of 82.05%, the third highest is by the ‘SGD’, ‘Ftrl’ and ‘AdaDelta’ optimizer.

6.0 ADVANTAGE AND TRADE-OFF

7.0 CONCLUSION

The "Parkinson's Disease Detection" dataset has provided significant findings and insights into the application of machine learning algorithms, using Feedforward Neural Networks (FNN) for the detection and monitoring of Parkinson's Disease (PD) in early detection. This research is aimed to implement machine learning to develop a non-invasive diagnostic tool that can identify Parkinson’s Disease through vocal characteristics, hence enhancing early diagnosis. This learning model can effectively classify individuals as either healthy or suffering from Parkinson's Disease (0 for healthy, 1 for PD) based on vocal features. The utilization of biomedical voice measurements from 31 individuals, of which 23 had Parkinson's, provided a structured dataset to train and validate the model. The best model achieved an accuracy of 92.31%, highlighting the potential of these techniques in practical diagnostic applications.

This research highlights the significance of hyperparameter adjustment in maximizing model performance by thorough experimentation with different hyperparameters. The learning rate of 0.002 and the number of epochs set at 159 were essential choices for obtaining maximum accuracy with a batch size of 15 as well. Different optimizers were tested and experimented as well, with Nadam having the highest accuracy of 92.31%, followed by Adam, Adamax, and RMSProp, each at 89.74%. The least effective was AdaDelta, with an accuracy of 76.92%. To prevent overfitting, dropout layers were implemented as a regularization strategy. Hence, the model maintained a balance between complexity and performance, allowing it to generalize well to new, unseen data. The activation functions used included ReLU for hidden layers and sigmoid for the output layer, both of which were effective for this classification task.

To conclude, not only can the model produce accurate diagnosis, it can also lighten the burden of healthcare officials. Healthcare professionals are able to concentrate more on treatment and care rather than detection. Continuous remote monitoring is another benefit of this research as it increases access to healthcare, particularly for those living in underserved or rural locations. Real-time data from the system enables customized treatment programs that are tailored to the individual requirements of every patient. Thereby, this can be concluded that Machine Learning can help automate the medical domain, as well as improving any country's healthcare.

8.0 REFERENCES

1. *UCI Machine Learning Repository*. (2019). Uci.edu. https://archive.ics.uci.edu/dataset/174/parkinsons
2. jaina. (2024). *Parkinson Disease Detection*. Kaggle.com. https://www.kaggle.com/datasets/jainaru/parkinson-disease-detection
3. Tran, C., Shen, K., Liu, K., Ashok, A., Ramirez-Zamora, A., Chen, J., Li, Y., & Fang, R. (2024). Deep learning predicts prevalent and incident Parkinson's disease from UK Biobank fundus imaging. *Scientific Reports*, *14*(1). https://doi.org/10.1038/s41598-024-54251-1
4. Clinic, C. (2022). *Parkinson's Disease: What It Is, Causes, Symptoms & Treatment*. Cleveland Clinic. https://my.clevelandclinic.org/health/diseases/8525-parkinsons-disease-an-overview

9.0 APPENDICES